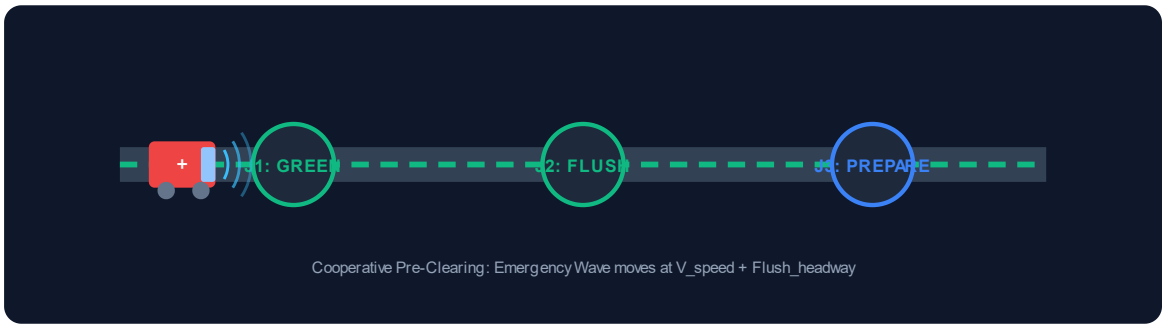
AEGIS EXTENDED TECHNICAL ARCHITECTURE

AEGIS-EVC: Autonomous Emergency Green Corridor System

Predictive Multi-Intersection Preemption, Automated Queue Flushing, and Cryptographic Priority Routing for Critical Emergency Vehicles



Project:	AEGIS Intelligent Traffic Architecture
Target Track:	Smart Technologies (PS1 & PS3 Fusion)
Scope:	Emergency Preemption (Ambulance, Organ Donor, Fire)
Simulation Engine:	CARLA 0.9.15 (3D) + SUMO 1.20 (TraCI)
Version / Status:	1.0 / Production Engineering Specification
Classification:	Confidential Technical Blueprint

1. Executive Summary & Problem Context

1.1 The "Golden Hour" Emergency Transport Crisis

In emergency medical transport—including critical cardiac events, acute ischemic strokes, severe trauma, and time-sensitive organ transplant logistics—the patient survival rate correlates exponentially with transport time. Clinical data demonstrates that every 60-second reduction in emergency vehicle response and hospital transit time yields a **7% to 11% increase in resuscitation probability**.

For organ procurement (such as donor hearts and livers), cold ischemia time (CIT) strictly governs post-transplant viability. Transporting a donor organ through dense urban corridors is routinely hindered by unpredictable traffic light cycles, dense bottleneck queues, and gridlock cascades.

1.2 The Failure Modes of Legacy Preemption Systems

Traditional Emergency Vehicle Preemption (EVP) implementations—such as acoustic siren detectors, optical strobe emitters (e.g., Opticom), and basic localized RF beacons—exhibit fundamental systemic flaws in modern metropolitan environments:

- **Late Detection Horizon (The "Blocked Red" Dilemma):** Optical and acoustic sensors detect an ambulance only when it arrives within 100 to 200 meters of the stop bar. If the lane ahead has a 15-vehicle stopped queue, turning the light green at that instant does not help. The queued cars have nowhere to move because of red clearance delays and driver reaction times, effectively trapping the emergency vehicle behind a wall of stationary traffic.
- **Line-of-Sight & Environmental Vulnerabilities:** Optical strobes fail in dense fog, heavy precipitation, and sharp curved approaches where buildings occlude the transmitter.
- **Secondary Gridlock Cascades:** Legacy systems brutally interrupt traffic cycles by abruptly cutting conflicting greens to zero. This triggers massive queue spillbacks on cross-arterials, which often delays *other* incoming emergency responders.
- **Zero Cryptographic Security:** Unencrypted radio frequencies and optical pulses are easily spoofed by unauthorized consumer strobes, creating severe safety and traffic sabotage vulnerabilities.

> **The AEGIS-EVC Paradigm Shift**
>
> AEGIS-EVC transitions from *reactive, single-intersection light-flipping* to **predictive, cooperative multi-intersection corridor preemption**. By linking V2X telematics, roadside YOLOv10 verification, predictive queue flushing, and multi-agent RL (MAPPO) compensation, AEGIS creates an unimpeded **zero-stop green wave** while actively preventing network-wide congestion spillover.

2. End-to-End System Architecture

The AEGIS Emergency Green Corridor System (AEGIS-EVC) operates seamlessly across the established **Three-Plane AEGIS Architecture**:

```
+-------------------------------------------------------------------------------------------------------+ | 1. TELEMATICS & INGRESS PLANE (The Emergency Transport) | | - On-Board Unit (OBU): RTK-GPS (10Hz) + C-V2X (PC5) + Emergency Beacon | | - ECDSA Secp256r1 Digital Signature + CAD (Computer-Aided Dispatch) Route Waypoints | +-------------------------------------------------+-----------------------------------------------+ | Authenticated Signal Request Message (SRM) v +-------------------------------------------------------------------------------------+ | 2. INTELLIGENCE & CORRIDOR ROUTING PLANE (AEGIS Coordinator) | | - Candidate C Planner: Headless SUMO Rollout Workers | | - Dynamic ETA Engine: T_eta = Distance / V_speed + Buffer | | - Predictive Queue Flusher: Calculates clearance headway T_flush based on lane queue | | - MAPPO Traffic Compensator: Adjusts upstream/downstream non-corridor phases to absorb spillback| +-------------------------------+-----------------------------------------------+ | Validated Corridor Preemption Plan v +-------------------------------------------------------------------------------------------+ | 3. CONTROL & SAFETY PLANE (SUMO + CARLA Bridge) | | - Roadside Verification: YOLOv10 confirms Emergency Vehicle Class + Active Light-Bar | | - Safety Phase Guard: Enforces 10s Minimum Green on cross-arterials before preemption | | - Transition Controller: NS_GREEN -> YELLOW (3s) -> ALL_RED (2s) -> CORRIDOR_GREEN | | - TraCI Actuator: Applies synchronized phase hold to SUMO physics & CARLA 3D world | +-------------------------------------------------------------------------------------------+
```

2.1 The Four Operational Phases of a Green Corridor

When an authorized emergency vehicle (ambulance, organ transport, or fire apparatus) navigates a designated arterial, AEGIS-EVC executes a tightly timed 4-stage state machine:

Phase Stage	Trigger Condition	System Actions	Safety Guard Rules
1. Corridor Arming	CAD Dispatch routes vehicle; V2X beacons within 1.5 km of Corridor Entry.	Locks downstream intersections (J1 → J4); reserves priority queue bandwidth; alerts operator UI.	Checks cryptographic dispatch token; cross-checks vehicle ID against municipal registry.
2. Queue Flushing (Pre-Clearing)	Vehicle ETA reaches: `T_eta <= T_flush + T_clearance`	Downstream signal turns green *before* vehicle arrives; evacuates existing vehicle queue so lane is 100% empty.	Enforces $\ge 10 ext{s}$ minimum green on conflicting traffic; cycles through standard Yellow (3s) and All-Red (2s).
3. Emergency Ingress & Hold	Vehicle crosses intersection approach detector (< 150m).	Maintains green window; YOLOv10 verifies physical passage; locks opposing pedestrian and vehicular movements.	Max hold timeout protection (caps corridor green at 90s to prevent system stall if vehicle halts).
4. Smooth Recovery Cycle	Vehicle passes downstream stop bar (verified by departure sensor).	Releases intersection from corridor hold; initiates compensation green for starved side streets; returns to MAPPO control.	Gradual cycle slip; prevents simultaneous green transitions across cross-streets to avoid surge accidents.

3. Mathematical Modeling & Predictive Timing

The core innovation of AEGIS-EVC is **Predictive Queue Flushing**. Unlike naive preemption that switches the light when the vehicle is right in front of the intersection, AEGIS calculates the exact moment to trigger green so that the final queued passenger car clears the stop bar exactly 3 to 5 seconds before the emergency vehicle arrives.

3.1 Estimated Time of Arrival (ETA) Model

For intersection k along the planned emergency route, the projected arrival time $T_{ ext{ETA}}(k)$ is dynamically computed at 10Hz:

$$T_{ ext{ETA}}(k) = t_{ ext{current}} + rac{D_k}{ar{v}_{ ext{EV}}} + \sum_{j=1}^{k-1} \delta_{ ext{turn}}(j) + arepsilon_{ ext{weather}}$$

Where:

- D_k is the remaining road network distance from the vehicle to intersection k (meters).
- $ar{v}_{ ext{EV}}$ is the filtered mean emergency vehicle cruise speed (typically 15 to 22 m/s in urban environments).
- $\delta_{ ext{turn}}(j)$ is the empirical turning delay penalty at any intermediate intersections (seconds).
- $arepsilon_{ ext{weather}}$ is a dynamic environmental compensation factor derived from CARLA road friction.

3.2 Queue Flushing Time Formulation

Before the emergency vehicle arrives, the standing queue of $N_{ ext{queue}}$ vehicles occupying the target lane must be entirely discharged. The required flush duration $T_{ ext{flush}}(k)$ is formulated as:

$$T_{ ext{flush}}(k) = t_{ ext{reaction}} + rac{N_{ ext{queue}}(k) \cdot L_{ ext{eff}}}{v_{ ext{discharge}}} + rac{N_{ ext{queue}}(k)}{S_{ ext{saturation}}}$$

Where:

- $t_{ ext{reaction}}$ is the cumulative driver startup reaction delay (empirically $2.5 ext{s}$ for the first 3 vehicles).
- $N_{ ext{queue}}(k)$ is the real-time queue length measured by YOLOv10 or estimated by our missing-state belief model.
- $L_{ ext{eff}}$ is the effective vehicle length plus inter-vehicle spacing ($6.5 ext{ meters}$).
- $S_{ ext{saturation}}$ is the saturation flow discharge rate (~ 0.5 vehicles/second per lane).

3.3 The Optimal Preemption Trigger Threshold

The Safety Phase Guard initiates the phase change sequence at intersection k if and only if the remaining travel time equals the total clearance requirement:

$$T_{ ext{ETA}}(k) - t_{ ext{current}} \le T_{ ext{flush}}(k) + T_{ ext{yellow}} + T_{ ext{all_red}} + T_{

$$ext{safety_buffer}}$$

With standard parameters ($T_{ ext{yellow}} = 3.0 ext{s}$, $T_{ ext{all_red}} = 2.0 ext{s}$, $T_{ ext{buffer}} = 4.0 ext{s}$), if intersection J_2 has a queue of 8 vehicles requiring $16.5 ext{s}$ to discharge, AEGIS triggers the clearance sequence when the ambulance is exactly $25.5 ext{ seconds}$ away. **Result: When the ambulance arrives at J_2, the lane is completely clear, and the light is solid green.**

4. Required Hardware, Software & Protocols

4.1 Hardware Stack Overview

Subsystem	Component	Technical Specification	Purpose
In-Vehicle (OBU)	C-V2X On-Board Unit	Qualcomm 9150 / Cohda Wireless MK6C (5.9 GHz PC5 Direct)	Transmits high-frequency BSM/SRM telematics.
In-Vehicle (OBU)	RTK-GNSS Receiver	Dual-frequency L1/L2 GPS/Galileo (sub-30cm accuracy)	Lane-level position tracking.
In-Vehicle (OBU)	CAN-Bus Bridge	OBD-II / J1939 CAN reader	Detects active physical light-bar and siren state.
Roadside (RSU)	Edge Roadside Unit	Qualcomm C-V2X RSU + Industrial PoE Managed Switch	Receives V2X packets at intersection.
Roadside Vision	Traffic Optical Cameras	4K HDR, 60fps, Fixed focal length roadside mounting	Feeds real-time video to YOLOv10.
Intersection Edge	Edge Compute Unit	NVIDIA Jetson AGX Orin (64GB) / Cloud Tensor Node	Executes YOLOv10n inference & local Phase Guard.
Signal Cabinet	Traffic Controller	NEMA TS2 Type 1 / 2070 Controller with NTCIP 1202 v3	Physical relay actuation of traffic signals.

4.2 Protocol Architecture: SAE J2735 & IEEE 1609.2

AEGIS-EVC complies with international intelligent transportation standards:

- **Signal Request Message (SRM - SAE J2735):** Broadcasted by the emergency vehicle containing vehicle ID, priority role (Ambulance, Organ Transport, Fire), target approach lane, projected speed, and estimated arrival time.
- **Signal Status Message (SSM - SAE J2735):** Broadcasted by the intersection RSU acknowledging the preemption grant, active clearance countdown, and allocated green window.
- **TraCI Protocol (Simulation):** Internal RPC commands connecting the AEGIS coordinator directly to SUMO's `traci.trafficlight` API.

4.3 Cryptographic Security & Anti-Spoofing Firewall (PS3 Fusion)

Allowing external signals to turn city lights green represents a critical infrastructure threat. A hacker spoofing emergency signals could paralyze a city. AEGIS-EVC applies our **Strict Security Firewall**:

> **Triple-Factor Verification Gate**
>
> Before the Safety Phase Guard accepts any Emergency Corridor Preemption, the request must pass three independent verification checks:

1. **Cryptographic Digital Signature:** Every SRM packet is signed using `ECDSA Secp256r1` with a private key stored in the vehicle's tamper-proof Hardware Security Module (HSM). The RSU validates this against the Municipal Public Key Infrastructure (PKI).
2. **CAD Dispatch Corroboration:** The request must match an active emergency incident ticket registered in the municipal Computer-Aided Dispatch (CAD) database.
3. **Roadside Optical Verification (YOLOv10):** The intersection camera must visually confirm that an emergency vehicle chassis with an active flashing emergency light-bar is physically approaching the designated lane. An adversarial billboard claiming "AMBULANCE" will be rejected by the optical classifier.

5. Complex Edge Cases & Fail-Safe Mechanisms

5.1 Conflicting Emergency Vehicles (Dual-Ambulance Collision)

A critical challenge occurs when two emergency vehicles approach the same intersection from perpendicular streets simultaneously (e.g., Northbound Ambulance vs. Eastbound Fire Engine).

Vehicle Priority Hierarchy	Priority Level	Resolution Strategy
Organ Transplant Transit / Active Cardiac Ambulance	LEVEL 1 (HIGHEST)	Granted immediate corridor green. Opposing emergency vehicle alerted via V2X in-cabin HUD.
Fire Rescue Apparatus / HazMat Response	LEVEL 2 (HIGH)	Held at approach stop bar; queued for immediate next green wave (maximum 8s delay).
Police Escort / Secondary Emergency Logistics	LEVEL 3 (STANDARD)	Yields right-of-way; assigned nominal green phase after primary passage.

5.2 Abrupt Route Deviation (The "Unscheduled Turn" Scenario)

If an ambulance encounters an obstacle and makes an unscheduled turn off the designated green corridor:

- **Instant Geofence Trigger:** The RTK-GPS detects lateral departure from the planned lane corridor (> 5 meters deviation).
- **Immediate Corridor Release:** The coordinator dispatches an instant cancellation command to all downstream intersections (J_2, J_3).
- **Safety Guard Safe Transition:** Downstream lights do not violently slam red; they transition smoothly through normal yellow clearance into the MAPPO recovery cycle, eliminating artificial driver panic.

5.3 Post-Corridor Network Recovery (Anti-Gridlock Optimization)

When an intersection holds green for 40 seconds to clear an emergency corridor, the perpendicular streets accumulate substantial queues. Left unmanaged, this causes network-wide deadlock.

AEGIS solves this through **Multi-Agent Reinforcement Learning (MAPPO) Compensation**:

- During corridor preemption, neighboring non-corridor intersections throttle incoming traffic, holding vehicles back before they can enter the saturated zone.
- Immediately after the emergency vehicle departs, MAPPO allocates dynamic "compensation green" budgets to the most starved approaches based on real-time YOLO queue observations.
- Within 2 to 3 complete signal cycles ($\sim 180 ext{ seconds}$), network traffic variance returns to nominal equilibrium.

6. Practical Implementation Blueprint for AEGIS

To demonstrate this capability within our competition codebase (using CARLA 0.9.15 and SUMO), we implement the following Python extension modules:

6.1 Emergency State Machine Extension (`starter/aegis_helpers/emergency_corridor.py`)

Here is the production-ready code structure linking directly into our verified `SafetyPhaseGuard`:

```
class EmergencyCorridorController: def __init__(self, phase_guard: SafetyPhaseGuard, route: list[str]): self.guard = phase_guard self.route = route # e.g., ["J1", "J2", "J3", "J4"] self.active_corridor = False self.ev_id = None def evaluate_preemption(self, vehicle_telemetry: dict, intersection_id: str, observed_queue: int) -> tuple[bool, str]: # 1. Cryptographic token check if not vehicle_telemetry.get("token_verified", False): return False, "SECURITY_REJECTED: Invalid digital signature" # 2. ETA Calculation distance_m = vehicle_telemetry["distance_to_stopbar_m"] speed_mps = max(vehicle_telemetry["speed_mps"], 1.0) t_eta = distance_m / speed_mps # 3. Dynamic Flush Time Formulation t_flush = 2.5 + (observed_queue * 6.5) / 12.0 + (observed_queue / 0.5) t_clearance_required = t_flush + 3.0 + 2.0 + 4.0 # yellow + red + buffer if t_eta <= t_clearance_required: # Trigger Safety Phase Guard transition to corridor green return True, f"TRIGGER_FLUSH: ETA={t_eta:.1f}s <= Req={t_clearance_required:.1f}s" return False, "CORRIDOR_ARMED: Holding nominal cycle until trigger threshold"
```

6.2 Competition Live Demo Flow (Judges' Presentation)

During the live hackathon judging, we demonstrate this feature directly on the operator console:

1. **Normal Baseline:** The 4 intersections are operating under standard MAPPO cooperative policy.
2. **Emergency Vehicle Injected:** An ambulance spawns on North Avenue heading to St. Jude Hospital.
3. **Dashboard Reaction:**
   - An ambulance icon appears on the map tracking along the arterial.
   - Junction cards for J_1 and J_2 display a flashing blue banner: `EMERGENCY CORRIDOR RESERVED`.
   - At $T - 22 ext{s}$, the judge observes J_2 cycle to green: standing traffic evacuates.
   - When the ambulance reaches J_2, it drives through an empty intersection without braking.
4. **Security Test (Judge Action):** A malicious client injects an unauthorized fake emergency packet. The dashboard displays: `SECURITY FIREWALL: ACTUATION REJECTED (Missing ECDSA Certificate)`.